

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 154

DELAYED-STAGING REAR-GUARD DRONE PROTOCOLS

Eyes astern, dropped on a velocity dip
laser thread home, tether back



the rear guard

STERN DROP · MMWAVE · LASER RELAY

NEVER FLY BLIND ASTERN

Defense system — v1.0 in force

GP-IM-154

Revision v1.0 — 23 August 2026

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VETTED BATCH 19 — PASS · STATION/COMMS/RECOVERY TESTS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 154

PHASE 154: DELAYED-STAGING REAR-GUARD DRONE PROTOCOLS — STATUS: CURRENT — PASS; STATION-KEEPING, COMMS AND RECOVERY PERFORMANCE TESTS OWED. The fleet has eyes behind. Phase 154 rejects the forward-only sensory layout — everything astern is a blind spot for debris, projectiles and hostile intercept vectors — and drops un-crewed Rear Shadow Guard Drones off the stern tracks during temporary velocity dips, so the escorts station-keep astern at secure distance without frame collisions. The drones sweep the aft quadrants with mmWave radar and lens arrays, beam telemetry home on Phase 17 line-of-sight laser links — no broadcast antennae, nothing to intercept — and when the mission clock runs out they sprint on side-thrusters to match velocity and glide into the rear hangar channels on Phase 20 tethers, no hot landing burns in the sub-decks. The vet graded the relative-velocity deployment sound; station-keeping, comms and recovery engineering are the owed tests, in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-154, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the drone law as seated (contents line, the three bodies — deceleration deployment, laser relays, recovery cycles — verbatim) — §2 why dropping back beats looking back (graded) — §3 the system astern — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 154: **Delayed-Staging Rear-Guard Drone Protocols.** Coordinates temporary forward velocity drops to safely eject un-crewed Rear Shadow Guard Drones from the stern tracks. The drones track perfectly astern at a secure multi-kilometer distance, utilizing secure line-of-sight laser networks to transmit 180-degree aft radar data back to the primary fleet.'

THE MASTHEAD LINE (verbatim): 'Industrial Defensive Strategy: Lagrangian Deceleration Staging & Trailing Laser Telemetry Relays [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Aft Sensory Blind Spots via Trailing Autonomous Escorts [extracted verbatim from the combined masthead line]'

THE LAGRANGIAN DECELERATION DEPLOYMENT (VERBATIM)

* Rationale: Rejects forward-only sensory layouts to eliminate fleet vulnerability to high-velocity trailing debris, kinetic projectiles, and non-natural hostile intercept vectors from astern. * Deceleration Drop Windows: Coordinates temporary forward velocity drops to safely eject un-crewed Rear Shadow Guard Drones from the stern tracks without physical frame collisions. * Calibrated Distance Stations: Main fleet acceleration leaves the drone tracking perfectly astern at a secure, multi-kilometer operational transmission distance along the flight path.

THE SECURE LINE-OF-SIGHT LASER RELAYS (VERBATIM)

* Wide-Aperture Aft Sweeps: Equips trailing escorts with high-frequency mmWave radar and lens arrays to execute continuous 180-degree monitoring of the aft tracking quadrants. * Antenna-Less Telemetry Feeds: Drone arrays beam real-time sensory data straight to the main ship's stern receivers via secure Phase 17 line-of-sight laser tracking networks. * Intercept Firebreak Buffers: Provides an early-warning window to allow real-time deployment of Phase 75 structural compression rolls or Phase 135 sacrificial mesh shields.

THE HIGH-VELOCITY RECOVERY & REFUEL CYCLES (VERBATIM)

* High-Output Speed Matching: Automated drone computers fire localized side-thrusters to rapidly accelerate and match primary fleet velocities upon mission telemetry completion. * Phase 20 Tether Stern Captures: Reclaiming drones glide into the rear hangar channels via high-tensile winch tethers, eliminating dangerous high-heat landing burns inside sub-decks. * Closed-Loop Maintenance Resets: Grained robotics hot-swap thruster fuel cassettes and deploy Phase 129 venturi scrubbers to re-rack the asset within a tight operational turnaround window.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE DEPLOYMENT, AFFIRMED: a temporary forward velocity drop lets the drone leave astern gently — relative velocity, not catapult force, does the separation; no collision risk to the frame, no launch impulse to budget. The slowest possible goodbye is the correct one for an escort that is meant to stay behind anyway.

THE LASER RELAY, AFFIRMED: line-of-sight laser telemetry is narrow, secure and effectively uninterceptable off-axis — the right bearer for a rear guard whose whole value is seeing without being seen. The phase's 'antenna-less' word is the correct instinct: no RF cone, no signature.

THE RECOVERY, AFFIRMED IN PRINCIPLE: sprint-to-match then tether-capture reuses the file's proven Phase 20 catch physics instead of reinventing docking. The honest bounds are the vet's: station-keeping propellant budgets, link geometry through the ship's own wake, and capture windows are engineering numbers — owed, in §9.

§3 — THE SYSTEM ASTERN

- THE DROP: temporary velocity dip; drones eject from the stern tracks — no frame contact.
- THE STATION: calibrated distance astern, tracking perfectly behind the fleet.
- THE EYES: high-frequency mmWave radar and lens arrays, 180-degree aft sweeps.
- THE LINK: Phase 17 line-of-sight laser telemetry to the stern receivers — antenna-less.
- THE RETURN: side-thruster sprint to match velocity; Phase 20 tether capture into the rear hangar channels; grained maintenance reset.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE REAR-GUARD DOCTRINE (seated with the phase): a fleet that only looks forward is a target with a blind hemisphere.
- THE QUIET-LINK DOCTRINE: the escort reports on a laser thread — seeing without broadcasting.
- THE FIREBREAK DOCTRINE: the drones are the early-warning buffer — an intercept meets a drone first.
- THE REUSE DOCTRINE: Phase 17 links and Phase 20 tethers — the rear guard is built from hardware the fleet already owns.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the drone: radar, lens arrays, laser transceiver, side-thrusters, tether hardpoint.
- STEP 2 — Bench the drop: ejection at a velocity dip — separation dynamics logged.
- STEP 3 — Bench the station: astern station-keeping propellant budget across a duty cycle.

- STEP 4 — Prove the link: laser lock through wake and maneuver; bit-rate and dropout maps.
- STEP 5 — Drill the recovery: sprint, match, tether capture into the hangar channel.
- STEP 6 — Publish the ledger: budgets, link maps, capture data — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 17 — the tracking networks: the laser relay bearer.
- PHASE 20 — the tether network: the recovery catch.
- PHASE 155 — the maglev brake (GP-IM-155): the astern family — the brake watches the front, the drones watch the back.
- PHASE 15 — the scout arrays: the forward perimeter this phase mirrors aft.

§8 — DO-NOT-BUILD

- DO NOT catapult the escort — the velocity dip is the deployment; impulse is collision risk.
- DO NOT broadcast — the laser thread is the link; an RF cone is a beacon.
- DO NOT burn into the hangar — tether capture only; hot exhaust in a sub-deck is the rejected failure.
- DO NOT fly the rear blind — the doctrine is the phase's name: the guard rides astern, always.

§9 — OWED

OWED: station-keeping, communications and recovery performance engineering — propellant budgets, link geometry through the wake, capture windows. Engineering tests, explicitly not physics objections. Seat additively when benched.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-154, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, seventh of the 20-chapter 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good

next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 154